

ALOR SETAR, Malaysia

WMO: 486030

Lat: 6.2005N

Lon: 100.4040E

Elev: 5

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 11	(b) 22.1	(c) 22.9	(d) 18.0	(e) 13.0	(f) 30.4	(g) 19.1	(h) 13.9	(i) 29.3	(j) 5.6	(k) 29.5	(l) 5.2	(m) 29.9	(n) 1.3	(o) 10	(p) 0.336

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	8.6	35.9	25.9	35.0	26.4	34.1	26.8	29.3	33.2	29.0	32.8	28.6	32.3	3.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	24.4	32.4	28.0	24.1	32.3	27.2	23.0	31.3	95.9	33.8	94.4	32.7	92.3	32.6	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 5.5	(b) 5.1	(c) 4.4	(d) 20.8	(e) 36.9	(f) 1.0	(g) 1.1	(h) 20.1	(i) 37.6	(j) 19.5	(k) 38.2	(l) 18.9	(m) 38.8	(n) 18.2	(o) 39.6	(p) 32.9	
			(q) 19.6	(r) 30.6	(s) 0.9	(t) 0.9	(u) 18.9	(v) 31.2	(w) 18.4	(x) 31.7	(y) 17.9	(z) 32.2	(aa) 17.2	(ab) 32.9	(ac) 32.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	28.0	28.8	29.4	29.4	29.1	28.7	28.2	28.2	28.2	28.1	28.1	27.8
	DBStd	1.12	0.99	1.02	1.00	0.97	0.93	0.92	0.97	1.00	1.03	0.93	0.98	0.97
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6754	558	528	603	582	592	561	564	544	544	562	543	553
	CDD18.3	3713	299	294	344	332	333	311	306	306	294	303	293	294
	CDH23.3	39543	3071	3276	3881	3818	3869	3548	3273	3185	2965	2972	2792	2892
	CDH26.7	16097	1264	1582	1895	1726	1637	1433	1220	1193	1074	1057	982	1034
(14) Wind	WSAvg	1.7	2.0	2.2	1.9	1.7	1.7	1.6	1.5	1.6	1.6	1.6	1.5	1.8
(15) Precipitation	PrecAvg	2067	33	48	110	197	233	165	189	203	298	295	210	89
	PrecMax	2530	258	322	201	483	439	388	324	352	634	434	352	252
	PrecMin	1430	0	0	1	38	91	53	66	43	18	135	44	3
	PrecStd	297	52	69	53	108	87	79	79	83	125	93	93	74
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	36.2	37.2	36.1	35.0	34.0	33.2	34.0	34.0	34.0	34.0	34.0
		MCWB	24.9	24.9	25.5	27.8	28.7	28.6	28.2	27.0	27.9	28.1	27.0	26.5
	2%	DB	33.9	35.2	36.1	35.0	34.0	33.1	32.9	33.1	33.0	33.0	33.1	33.0
		MCWB	25.4	25.5	25.8	27.8	28.6	28.2	28.0	27.3	27.5	27.7	27.0	26.4
	5%	DB	33.0	34.2	35.1	34.1	33.1	32.8	32.1	32.2	32.2	32.1	32.2	32.2
		MCWB	25.4	25.5	26.0	27.9	28.2	28.1	27.7	27.2	27.1	27.3	26.9	26.2
	10%	DB	32.1	33.8	34.1	33.1	32.2	32.0	31.8	31.9	31.8	31.2	31.8	31.2
		MCWB	25.3	25.5	26.3	27.8	27.8	27.6	27.6	27.2	27.1	27.1	26.9	26.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	28.3	29.1	29.9	29.9	29.6	29.2	29.0	28.8	29.1	29.3	28.9
		MCDB	32.2	33.8	34.6	34.5	33.7	32.6	32.3	32.5	32.7	32.7	32.3	32.4
	2%	WB	27.4	27.5	28.3	29.2	29.2	28.9	28.7	28.2	28.1	28.4	28.4	28.0
		MCDB	31.5	32.9	33.4	33.6	33.0	32.3	31.8	31.7	31.8	31.9	31.7	31.4
	5%	WB	26.6	26.8	27.6	28.6	28.7	28.4	28.1	27.9	27.6	27.8	27.7	27.1
		MCDB	30.8	32.1	32.5	32.9	32.3	31.5	31.1	31.3	31.1	31.3	31.0	30.8
	10%	WB	26.0	26.3	27.0	28.1	28.1	27.9	27.7	27.3	27.1	27.2	26.9	26.4
		MCDB	30.1	31.5	31.9	32.1	31.5	30.9	30.7	30.5	30.5	30.5	30.2	30.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.6	10.4	8.6	7.4	7.0	7.0	7.4	7.3	7.3	7.5	7.5
		MCDBR	10.1	11.4	11.9	9.5	8.1	7.6	7.7	8.0	8.1	8.2	8.4	8.8
		MCWBR	3.9	3.9	4.3	4.3	4.2	4.1	4.2	3.9	3.9	4.1	3.9	3.8
	5% WB	MCDBR	8.8	10.5	10.2	8.8	7.8	7.2	7.3	7.6	7.7	7.9	8.1	8.1
		MCWBR	4.4	4.5	4.7	4.6	4.4	4.2	4.4	4.2	4.0	4.4	4.4	4.3
(40) Clear-Sky Solar Irradiance	taub		0.438	0.459	0.470	0.475	0.460	0.472	0.474	0.476	0.468	0.461	0.442	0.437
	taud		2.362	2.277	2.269	2.263	2.327	2.317	2.298	2.279	2.301	2.318	2.373	2.385
	Ebn at Noon		880	870	860	841	835	814	816	829	849	857	870	874
	Edh at Noon		126	140	142	141	128	127	131	137	135	133	123	121
(44) All-Sky Solar Radiation	RadAvg		5.02	5.86	5.84	5.69	5.13	4.97	4.86	4.92	4.77	4.63	4.44	4.34
	RadStd		0.35	0.36	0.49	0.24	0.26	0.21	0.21	0.20	0.21	0.33	0.28	0.38

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.22	+0.14	N/A	+0.32	+0.22	+0.19	N/A	N/A	+94	+94

Nomenclature: See separate page